MIL-C-28546A(YD)
29 November 1976
SUPERSEDING
MIL-C-28546(YD)
31 May 1968

MILITARY SPECIFICATION

CRANES, OVERHEAD TRAVELING, UNDERHUNG, ELECTRIC POWERED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE, DEFINITIONS, AND CLASSIFICATIONS

* 1.1 Scope. This specification covers overhead traveling electric powered cranes of both single and double girder, and runway track systems of the underrunning type, with capacities up to 10 tons. Overhead electric traveling (OET) cranes covered in this specification are not intended for heavy or severe duty use; for use in handling hot metal or fissionable material; or for use in hazardous areas as defined in National Fire Protection Association (NFPA) No. 70.

* 1.1.1 Definitions. To ensure clarity of understanding of the applicability of this specification the below selected definitions of terms used herein are provided:

"Overhead Electric Traveling (OET) Crane". Any electrically powered crane with a movable bridge that travels on either the lower flanges ("underhung" or "underrunning" type) or the top surface of rails ("top running" type) of a fixed runway structure located above ground level.

"Underhung (underrunning) Crane". A specific type of OET that runs on the lower flanges of an overhead fixed runway structure and carries a movable or fixed hoisting mechanism.

* 1.2 Classification. OET cranes shall be of the following types and service classes, as specified (see 6.2.1):

Type I - Single girder.
Type II - Double girder.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Class A1 - Standby service.
Class A2 - Infrequent service.
Class B - Light service.
Class C - Moderate service.

(The service classes listed above roughly correspond to like-lettered classifications given in Crane Manufacturers Association of America (CMAA) No. 70 or 74. However, the design requirements specified herein differ in many respects from the design requirements of CMAA No. 70 or 74.)

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-H-19925 - Hoists, Wire Rope, Electric Powered.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

Gear Standards.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, N.W., Washington, DC 20005.)

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Design, Fabrication & Erection of Structural Steel of Buildings.

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 101 Park Avenue, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

B30.2 - Safety Standard for Overhead and Gantry Cranes.
B30.11 - Safety Standards for Monorail Systems and Underruning Cranes.
B30.16 - Safety Standards for Overhead Hoists.
B30.17 - Safety Standards for Single Girder Top Running Cranes.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A27 - Specification for Mild- to Medium-Strength Carbon-Steel Castings for General Application.
A36 - Specification for Structural Steel.
A53 - Welded and Seamless Steel Pipe.
A108 - Cold-Finished Carbon Steel Bars and Shafting.
A148 - Specification for High-Strength Steel Castings for Structural Purposes.
A235 - Specification for Carbon Steel Forgings for General Industrial Use.
A236 - Specification for Carbon Steel Forgings for Railway Use.
A237 - Specification for Alloy Steel Forgings for General Industrial Use.
A325 - High-Strength Bolts for Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers.
A434 - Quenched and Tempered Alloy Steel Bars, Hot-Rolled or Cold-Finished.
A439 - Specification for Austenitic Ductil Iron Castings.

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A576 - Special Quality Hot-Rolled Carbon Steel Bars.
B148 - Specification for Aluminum Bronze Sand Castings.
B584 - Specification for Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CRANE MANUFACTURERS ASSOCIATION OF AMERICA, INC. (CMAA)

No. 70 - Specifications for Electric Overhead Traveling Cranes.
No. 74 - Specifications for Toprunning & Underrunning Single Girder Electric Overhead Traveling Cranes.

(Application for copies should be addressed to the Crane Manufacturers Association of America, Inc., 1326 Freeport Road, Pittsburgh, PA 15238.)

HOIST MANUFACTURERS INSTITUTE (HMI)

HMI 100 - Electric Wire Rope Hoists.

(Application for copies should be addressed to the Hoist Manufacturers Institute, 1326 Freeport Road, Pittsburgh, PA 15238.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems.
MG1 - Motors and Generators.
WC3 - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J551 - Standard for Measurement of Electromagnetic Radiation from Motor Vehicles (20-1000 MHz).
J552 - Recommended Practice for Radio Noise Suppressors.
J534 - Standard for Lubrication Fittings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Description. The OET cranes furnished under this specification shall be the crane manufacturer's standard product as furnished to the commercial market and shall conform to CMAA No. 70 or 74, except as specified herein. Each crane service class in this specification corresponds to the service class of the same designation in CMAA No. 70 or 74. Underhung electric powered overhead traveling cranes shall consist of a complete traveling hoist and bridge assembly with end trucks, current collectors, and controls ready for immediate operation when assembled and placed on a runway, as detailed in the schematic arrangement drawings (see 6.7). The cranes shall be of the single or double girder type, indoor or outdoor type, cab or pendant controlled, and shall have the spans, capacities, speeds, and lifts as specified.

* 3.2 Standard commercial product. The cranes shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the cranes being furnished. A standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model(s).

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* 3.3 Safety. Each crane furnished shall conform to the applicable mandatory and advisory safety requirements of ANSI B30.2, B30.11, B30.16, and/or B30.17 and HMI 100, except as modified herein.

* 3.4 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purposed in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly. All nuts, bolts, and screws shall have standard screw threads in accordance with NBS H28.

* 3.4.1 Structural steel. Structural steel shall conform to ASTM A36. The minimum thickness of material shall be 0.3125 inch except that the webs of channels and beams, and resistor frames may be 0.25 inch, and sheet steel for cabs and other enclosures may be 0.10 inch.

* 3.4.2 Cast steel. Cast steel shall conform to ASTM A27, grade 60-30, or A148, as required for design limitations.

* 3.4.3 Forged steel. Forged steel shall conform to ASTM A235, A236, or A237, class as applicable.

* 3.4.4 Seamless steel pipe. Seamless steel pipe shall conform to ASTM A53, type S, grade B, and shall be of the open hearth or basic oxygen steel.

* 3.4.5 Hot-rolled steel bars. Hot-rolled steel bars, up to and including 5-inch diameter stock, shall conform to ASTM A576, grade designation as applicable, or to ASTM A434, class as applicable. Forged steel may be used in lieu of hot-rolled steel bars.

* 3.4.6 Cold-rolled or cold-drawn steel bars. Cold-rolled or cold-drawn steel bars shall conform to ASTM A108, grade designation as applicable.

* 3.4.7 Cast iron. Cast iron shall conform to ASTM A27, grade not less than 60-30, or ASTM A148, grade as applicable.

* 3.4.8 Nodular graphitic cast iron. Nodular graphitic cast iron shall conform to ASTM A439, type as applicable to the design tensile and yeild strength, and the Brinell hardness required.

* 3.4.9 High strength bolts. High strength bolts for structural steel joints shall conform to ASTM A325.

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* 3.4.10 Bronze. Bronze shall conform to ASTM B584, alloy number as applicable. Aluminum bronze shall conform to ASTM B148, alloy number 9C.

* 3.5 Repair parts service. Crane manufacturer shall guarantee the availability of repair parts for the crane furnished, for a period of not less than 10 years. Repair parts shall be readily available when ordered by part number taken from the manufacturer's standard repair parts book supplied with the crane.

3.6 Capacity. Capacity rating of the cranes shall be in tons, of 2,000 pounds, and shall be indicated on the capacity plate attached to the crane. The capacity shall be as specified (see 6.2.1). All handling accessories, such as slings and spreader bars, shall be considered as part of the load to be lifted, but the hook and the block shall not.

* 3.7 Performance. When operated in accordance with the safety rules of ANSI B30.2, B30.11, B30.16, and/or B30.17, and HMI 100, through a representative repetitive cycle of motions and loads commensurate with the specified crane service class, no electrical part shall sustain a temperature rise in excess of the applicable allowable limits of NEMA MG1 and ICS for the corresponding part or item. Steady running rated travel, lifting, and lowering speeds developed under test with full-rated load shall not vary from specified speeds by more than $\pm$10 percent for direct current and alternating current operated cranes.

* 3.8 Structural design. Structural design shall conform to applicable portions of CMAA No. 70 or 74 and AISC except as specified herein.

* 3.8.1 Stresses. Load carrying parts, except structural members and hoisting ropes, shall be designed so that the calculated static stress in the material, based on rated load, shall not exceed 20 percent of the assumed average ultimate strength of the material.

* 3.8.2 Wind load. Unless otherwise specified (see 6.2.1), wind load for outdoor cranes shall be considered as 5 pounds per square foot (psf) from any direction to the projected area.

3.8.3 Girders. The crane girders shall be welded structural steel box sections, wide flange beams, standard "I" beams, reinforced beams, or box sections, fabricated from structural shapes. The manufacturer shall specify the type and construction of the girders to be furnished.

3.8.4 Footwalks and handrails. When specified (see 6.2.1), a footwalk and handrail shall be provided. The handrail shall be at least 42 inches high and provided with an intermediate railing. The footwalk may be wood, anti-slip plate, expanded metal or grating as specified by the crane manufacturer. It shall be protected on all exposed edges by a suitable toe guard.

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* 3.8.5 Operator's cab. Unless otherwise specified (see 6.2.1), cab controlled cranes shall have the cab located at one end of the crane bridge. The cab entrance shall mate with a building mounted crane access platform where detailed on the schematic arrangement drawings (see 6.7). Unless otherwise specified (see 6.2.1), indoor cab controlled cranes shall have open type cabs. Unless otherwise specified (see 6.2.1), all cabs shall be equipped with a comfortable operator's seat and a loud gong or horn type warning device, operated by a momentary contact pushbutton or foot operated controller.

* 3.8.6 Transfer point controls. Control of transfer points between crange bridges, and between crane bridges and monorail systems, shall be by means of chains or nylon ropes as specified (see 6.2.1). When specified (see 6.2.1), control shall be by means of electric motors, controlled by pendant or fixed pushbutton stations as indicated on the schematic drawings.

* 3.8.7 Track beam suspension. Unless otherwise specified (see 6.2.1), runway beam suspension shall be of the flexible or rigid type at the manufacturer's option. The span of the truck shall be as specified (see 6.2.1).

3.8.8 Track beam. Runway beams shall conform to the requirements of AISC.

* 3.9 Mechanical design. Mechanical design shall conform to CMAA No. 70 or 74 except as specified herein.

* 3.9.1 Trolley and hoist. The trolley and hoist shall conform to the requirements of MIL-H-19925, type I, drive M of the same class as the bridge drive unless otherwise specified (see 6.2.1). The trolley and hoist capacity, type, lift, trolley travel speed, and lift travel speed shall be as specified (see 6.2.1).

* 3.9.2 Gearing. All gears and pinions shall be of steel, except worm gears may be of high strength bronze. Gearing shall be straight spur or bevel, parallel shaft helical, herringbone, or spiral bevel type for all cranes, except that worm gearing may be used on trolley drives for class A1, A2, and B cranes and shall be in accordance with AGMA criteria.

* 3.9.3 Bearings. All bearings shall be of the antifriction type, except that bearings subject to a slight rocker motion, such as the idler sheave bearing, may be of the bronze bushing type. The B-10 life rating of anitfriction bearings shall be 3,000 hours for class A1, A2, and B cranes and 5,000 hours for class C cranes.

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* 3.9.4 Bridge drives. Bridge drives shall be electric powered. Unless otherwise specified (see 6.2.1), bridge drive arrangement shall be selected by the manufacturer and shall conform to CMAA No. 70 or 74. Bridge travel speed shall be as specified (see 6.2.1).

* 3.9.5 Couplings. All couplings shall be located immediately adjacent to a bearing. Couplings between closely spaced bearings shall be of the full flexible type. Couplings for use with floating shafts shall be of the half flexible type with pilots. Rigid couplings shall be steel or malleable iron and shall be of the compression or safety flange type.

* 3.9.6 Wheels. Wheels and wheel loading shall be in accordance with CMAA No. 70 or 74, unless otherwise specified (see 6.2.1). When rotating axles are used, drive wheels shall be press fitted and keyed to the axles, and idler wheels shall be press fitted.

* 3.9.7 Stops. All cranes shall be provided with stops in accordance with the provisions of the appropriate ANSI standards.

* 3.10 Electrical equipment. The electrical equipment shall conform to NFPA No. 70 and CMAA No. 70 or 74, except as specified herein, and shall be designed to operate from the voltage, frequency, and phase specified (see 6.2.1). The design for wiring, insulation, allowable voltage drop, current collectors, control, overcurrent protection, and grounding shall be in accordance with NFPA No. 70, as applicable, and as specified herein.

* 3.10.1 Motors. Each crane motion (bridge, trolley, and hoist) shall be driven by a separate motor (or motors, for duplex bridge drive) independent of other motions. Hoist and trolley shall conform to MIL-H-19925. Bridge motors shall be suitable for the service class specified for the crane (see 1.2). Bridge motors shall conform to NEMA MG1 and the following:

(a) Motor mounting, shaft, and key dimensions shall conform to the manufacturer's standard.
(b) Temperature rise shall be used in accordance with NEMA MG1, for the motor type and insulation used.
(c) Enclosure: totally enclosed.
(d) Maximum rated speed: 1,800 revolutions per minute (rpm) (nominal).
(e) Bearings: ball or roller.
(f) Insulation: NEMA MG1, part 1, class A, B, F, or H.
(g) Bridge motor starting torque shall be a minimum of 160 percent of full load torque.

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(h) Time ratings: Classes A1, A2, B and C - 30 minutes. The crane manufacturers shall ensure that the motors are certified to be at not less than 30 minutes by the motor manufacturer in accordance with NEMA MG1, as applicable.

* 3.10.2 Controllers. Controllers shall be provided with NEMA rated or definite purpose contactor specifically rated for crane service. The magnetic contactors used shall not exceed the manufacturers published ratings. Controllers shall be mounted in enclosures conforming to NEMA ICS, part 1-110, type as applicable to the hoist characteristics and use. When resistors are mounted in the same enclosure as controllers, air circulation by natural convection or forced ventilation shall be provided. Bridge controls shall conform to NEMA ICS, class 1 or 2 as specified (see 6.2.1). Hoist and trolley controls shall conform to MIL-H-19925, as specified (see 6.2.1). When specified (see 6.2.1), hook shall be capable of lifting and lowering rated load with adjustable speed range between 1 foot per minute (fpm) and 15 fpm.

* 3.10.2.1 Speed control NEMA, class 1. Speed control class 1 shall provide reversing single or two-speed control of a single or two-speed squirrel-cage motor for bridge travel service only, as specified (see 6.2.1). When specified (see 6.2.1), electronic, silcon controlled rectifier (SCR), type controls, or primary resistor reduced voltage starting, acceleration, and deceleration shall be provided for bridge service regardless of the motor horsepower.

* 3.10.2.2 Speed control NEMA, class 2. Speed control class 2 shall provide reversing, plugging, five-step, secondary-resistance control of a wound rotor motor for bridge travel service only. The first running point shall provide not more than 50 percent of rated torque during starting.

* 3.10.3 Enclosures. Enclosures for control panels, controls, and brakes shall be as specified (see 6.2.1).

* 3.10.4 Resistors. Resistors, when furnished, shall be in accordance with the applicable class ratings as specified in NEMA ICS.

* 3.10.5 Protection. Protection shall be in accordance with CMAA No. 70 or 74 and NFPA No. 70.

* 3.10.6 Heat shields. When specified (see 6.2.1), heat shields shall be provided to protect electrical components from high radiant heat.

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* 3.10.7 Pendant pushbuttom stations. Pendant pushbuttons for floor controlled cranes shall be legibly and permanently marked in order, in a single vertical line from top to bottom in accordance with MIL-H-19925. The length of the pendant control, measured as the distance from the underside of the track beam to the top of the pushbutton station, shall be as specified (see 6.2.1).

* 3.10.8 Bridge electrification system. Each bridge electrification system shall be of the enclosed safety or festooned type.

* 3.10.8.1 Enclosed safety type. Enclosed conductors shall be rigid shapes of galvanized steel, copper clad steel, or copper. Supports shall be spaced at intervals of not more than 80 times the vertical dimensions of the conductors, but in no case greater than 15 feet apart. All supports, brackets, insulators, fastening devices, conductors, and power takeoffs necessary for complete installation shall be furnished. Conductors shall be covered with snap-on or slip-on plastic insulation (including end and joint covers) so shaped as to prevent accidental contact with the line conductors. Collectors shall be double-wheel, or sliding-shoe type and shall match their respective electrification systems. Collector arms shall be insulated, spring-loaded, and shall permit sufficient lateral and vertical movement to allow for imperfect track, and misalinement of the collector support in relation to the conductors.

* 3.10.8.2 Festooned type. Festooned type conductors shall be rope-stranded, flexible, insulated, portable cable and conforming to NEMA WC3. Cables shall be supported wheeled carriers running in a messenger track. The carriers and messenger track shall form an integrated system manufactured specifically for the use intended, and shall function in a manner which protects the conductor cables from abrasion, kinking, or excessive twisting. Cables shall be coiled against the lay of cable stranding so that the cable relaxes as it is extended.

* 3.10.9 Runway electrification system. When specified (see 6.2.1), the crane manufacturer shall provide the entire runway electrification system, which shall be of the enclosed safety type as specified herein (see 3.10.8).

* 3.11 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag or spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable, after cleaning, and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The

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painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be as specified (see 6.2.1).

* 3.12 Technical data. When specified (see 6.2.1), prior to commencing the fabrication of any work, the contractor shall forward design drawings and data to the contracting officer for approval in accordance with the contract. The design drawings (see 6.3) shall show complete calculations for, and shall define the applicability of design loads, moments, shears, deflections, stresses, torques, rating speeds, life of bearings, rope leads, material and metal hardness characteristics, and factors of safety, as applicable, for the structural and mechanical elements of the crane.

* 3.13 Technical manuals. When specified (see 6.2.1), complete operating and maintenance instructions and parts list (see 6.3) shall be forwarded in accordance with the contract.

* 3.14 Supervisory erector services. When specified (see 6.2.1), the crane contractor shall provide the services of a supervisor who shall be responsible for directing the unpacking and erection of the equipment. The supervisor shall also witness and assist in the performance of quality assurance tests.

* 3.15 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

* 3.16 Instruction plates. A complete electrical schematic diagram, complete lubrication chart, and safety warnings shall be furnished and installed in a location which facilitates use of the charts and diagrams and accurately provides the intended warning. They may be encapsulated plastic, etched plastic or metal, or engraved plastic or metal, and shall be legible and durable.

* 3.17 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

* 3.18 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. All parts requiring lubrication shall be lubricated prior to delivery.

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* 3.18.1 Lubrication fittings. Hydraulic lubrication fittings shall be in accordance with SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts.

* 3.19 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.20 Radio noise suppressors. When specified (see 6.2.1), cranes shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the crane shall be within the limits of SAE J551.

* 3.21 Identification plates. Unless otherwise specified (see 6.2.1), identification plates shall be furnished by the manufacturer. They shall conform to MIL-P-514, type I, style 1, composition C. Plates shall be affixed to the crane in a conspicuous place with brass screws or bolts of not less than 1/8-inch diameter.

* 3.22 Atmospheric conditions. Unless otherwise specified (see 6.2.1), all classes of cranes shall be designed to operate in normal ambient temperatures of 0° Fahrenheit (F) to 100°F, and in atmospheric conditions free from excessive dust, moisture, and corrosive fumes.

* 3.23 Field erection. When specified (see 6.2.1), the complete system shall be field erected, alined, painted, adjusted, and connected to the power source ready for testing in accordance with 4.3.

3.24 Workmanship.

* 3.24.1 Steel fabrication. The steel used in fabrication shall be free fron kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.24.2 Bolted connections. All bolt holes shall be true and accurate and shall have the burrs removed. Washers or lockwashers shall be as provided and as required in accordance with standard commercial practices. All nuts, bolts, and screws shall be tight.

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* 3.24.3 Riveted connections. All rivet holes shall be true and accurate and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be an approved shape and shall be concentric with the rivet. The rivet head shall make full contact with the intended surface. Rivets shall be so fabricated as to develop a joint strength of not less than the design value.

* 3.24.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with CMAA No. 70 to 74. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

* 3.24.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

* 3.25 Shop assembly. When specified (see 6.2.1), each crane shall be assembled, except for the rope reeving, at the contractor's plant, wired, examined, and tested as specified (see 4.2 and 4.3.1). Cabs shall be completely assembled with all glazing, equipment, conduit, wiring, and other parts. For test purposes, completely assembled cabs need not be mounted on the bridge or trolley and may be electrically connected to the crane with temporary wiring. Hydraulic lines for brakes need not be assembled in the shop.

* 3.26 Disassembly for shipment. Crane disassembly for shipment shall be the minimum practicable. The bridge girders shall be shipped full length with as much of the bridge drive and footwalks assembled as physically practicable. The end trucks and trolley(s) shall be shipped completely assembled. The cab shall be shipped as a complete unit. Parts which are disassembled for shipment shall be matchmarked and tagged with metal tags. The tags and markings thereon shall be waterproofed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Examination. The crane shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements, or presence of one or more defects preventing or lessening maximum efficiency, shall constitute cause for rejection.

4.3 Tests.

* 4.3.1 Shop tests. Each crane, after being assembled as specified (see 3.25 and 6.2.1), at the contractor's plant, shall be tested without load to determine successful operation of the assembled unit. The assembled crane shall be square and in alinement with parts fitted and adjusted properly. Operation shall be smooth and quiet. Bearings shall be cool running. Motor controls, as applicable, shall be timed and adjusted to give smooth acceleration and minimum current peaks. After completion of motor control adjustments, the protective equipment shall be set to function at slightly higher than normal peak current and slightly lower than normal low voltage.

* 4.3.2 Operating tests. The bridge and trolley brakes shall be adjusted by the contractor so as to stop the bridge and trolley without excessive swinging of the hoist ropes. After the completed cranes are erected in place, adjusted, lubricated, and otherwise made ready for operation, they will be tested by the Government, in the presence of the contractor, as indicated on Test Forms A through H inclusive, which are attached to this specification. Bridge and trolley travel and hoisting tests for speed will be made after mechanisms have attained essentially steady state, prior to taking data. The Government will furnish the labor, power, slings, and weights required for the operating tests. Tests will be conducted observing the safety rules of applicable ANSI standards.

* 4.3.3 Measurement of electromagnetic radiation. When electromagnetic interference characteristics are required, the equipment shall be measured in accordance with SAE J551 to determine conformance to 3.20. When suppressed to conform to 3.20, the manufacturer may furnish a certification (see 6.3), that the crane meets the requirements in lieu of the test, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.4 Packaging inspection. The preservation-packaging, and packing of the cranes shall be inspected to verify conformance to the requirements of Section 5.

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5. PACKAGING

* 5.1 Preservation-packaging, and packing. Unless otherwise specified (see 6.2.1), the complete crane shall be preserved and packaged in accordance with the manufacturer's standard practice, shall be packed in a manner which will insure arrival in satisfactory condition, and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules. Cranes for overseas destinations shall be preserved, packaged, and packed as specified (see 6.2.1).

6. NOTES

* 6.1 Intended use. OET cranes are used in shop, maintenance, assembly, repair, and storage areas. OET cranes covered by this specification are intended to be procured as a part of a construction project covered by a project specification. OET cranes are used in building where the floor work area must be completely free from posts and structural obstructions. Allowable options (when specified, unless otherwise specified, and as specified) in this specification will be furnished as detailed in the project specification.

6.2 Ordering data. Procurement documents should specify:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type and service class of crane required (see 1.2).
(c) The crane capacity (see 3.6).
(d) When wind load, in psf, is different from that specified in 3.8.2.
(e) When footwalks and handrails are required (see 3.8.4).
(f) When the cab location is different than as specified in 3.8.5.
(g) When the indoor cab shall not be an open type (see 3.8.5).
(h) When an operator's seat or warning device is not required in cab controlled cranes (see 3.8.5).
(i) Whether control of transfer point interlocks shall be by means of chains, nylon ropes, or by electric motor controlled by pendant or fixed control station operation (see 3.8.6).
(j) When track beam suspension shall not be of the flexible or rigid type (see 3.8.7).
(k) The span of the track beams (see 3.8.7).
(l) When trolley and hoist shall not conform to MIL-H-19925 (see 3.9.1).

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(m) Trolley and hoist capacity, type, lift, trolley travel speed, and lift travel speed in accordance with MIL-H-19925 (see 3.9.1).
(n) When bridge drive arrangement shall be other than that specified (see 3.9.4).
(o) The bridge travel speed (see 3.9.4).
(p) When wheels shall not conform to CMAA No. 70 or 74 (see 3.9.6).
(q) Voltage, frequency, and phase of electrical current (see 3.10).
(r) Speed control class 1 or 2 for bridge controls (see 3.10.2).
(s) Hoist and trolley speed control class in accordance with MIL-H-19925 (see 3.10.2).
(t) When hook lifting and lowering speeds shall be as specified (see 3.10.2).
(u) When speed control class 1 shall be single or two speed (see 3.10.2.1).
(v) When electronic SCR type controls, or reduced voltage starting, acceleration, and deceleration are required (see 3.10.2.1).
(w) Type of enclosure for control panels, controls, and brakes, (see 3.10.3).
(x) When heat shields are required (see 3.10.6).
(y) The length of the pendant control (see 3.10.7).
(z) When the entire runway electrification system is to be furnished (see 3.10.9).
(aa) Color of paint (see 3.11).
(bb) When technical data is required (see 3.12).
(cc) When technical manuals are required (see 3.13).
(dd) When supervisory erector services are required (see 3.14).
(ee) When fungus proofing is required (see 3.19).
(ff) When radio noise suppression is required (see 3.20).
(gg) When identification plates are to be furnished by the contracting officer (see 3.21).
(hh) When temperature and atmospheric conditions are different than as specified (see 3.22).
(ii) When crane is to be field erected (see 3.23).
(jj) When shop assembly and tests are required (see 3.25 and 4.3.1).
(kk) When preservation-packaging, and packing is different from that specified in 5.1 and the requirements for overseas destinations.

* 6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not

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invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DD 1664
3.12	Drawings, Design Evaluation	UDI-E-24001A(YD)
3.13	Manuals, Technical Modified Commercial	UDI-M-24006C
4.3.3	Certificate of Compliance	DI-E-2121

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 6.4 Service class selection. The following discription of crane usage and duty cycle should serve as a guide in selecting the proper crane service class.

Class A1 - standby service. This service class covers cranes used in installations such as: power houses, turbine rooms, nuclear reactor buildings, motor rooms, nuclear fuel handling and transformer stations, where precise handling of valuable machinery at slow speed with long idle periods between lifts is required. Capacity loads may be handled for initial installation of machinery and for infrequent maintenance.

Class A2 - infrequent service. These cranes will be used in installations such as: small maintenance shops, pump rooms, testing laboratories, and similar operations where the loads are relatively light, the speeds are slow, and a low degree of control accuracy is required. The loads may vary anywhere from no load to full capacity with a frequency of a few lifts per day or month.

Class B - light service. This service covers cranes, such as used in repair shops, light assembly operations, service buildings, and light warehousing where service requirements are light and the speed is slow. Loads may vary from no load to full rated load with an average load of 50 percent of capacity with two to five lifts per hour, averaging 15 feet.

Class C - moderate service. This service covers cranes, such as used in machine shops, fabricating and assembly or overhaul shops, and warehouse work where the service requirements are medium. In this type of service the crane will handle loads which average 50 percent of the rated capacity with five to ten lifts per hour, averaging 15 feet.

* 6.5 Speeds. Suggested operating speeds for hoists and trolleys are contained in a table in MIL-H-19925. For cab controlled cranes, unless usage dictates otherwise, selected bridge speeds should be in the 150 to 300 fpm range and for floor controlled cranes, bridge speeds should be in the 75 to 200 fpm range.

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* 6.6 Selection of speed control classes. Selection of speed control classes should be based primarily on the ability of the load to withstand starting acceleration, deceleration, and stopping forces. The following handling characteristics can be expected from the indicated speed control classes:

Handling characteristics	Motion	
	Hoist	Bridge and trolley
Smooth acceleration, broad speed range.	---	Class 2
Medium acceleration, fixed speeds.	---	Class 1
Rapid acceleration, fixed speeds.	Class 1	Class 1

Cost is frequently a consideration in selection of speed control classes. The higher the number of speed control class, the higher will be the cost.

* 6.7 Schematic drawing. The project specification should include a schematic drawing or sketches showing the following, as well as other information that may seem appropriate:

(a) Location and orientation, including the cardinal points, where the crane complex is to be erected, together with floor plan and layout of runways.
(b) Span, runway track beam, and electrical contact conductor detail of rails on which cranes consisting of a hoist, trolley, bridge, and end truck assembly shall operate.
(c) Complete details concerning any existing construction of which the new crane complex construction shall become a part.
(d) Detailed location of service platforms; pits to be served; and lift ranges, capacities, duty classes, and speeds of all hoists; and any special lifting devices required.
(e) Available supports, columns, and building structural members which may be used for support or suspended runway track beams, together with maximum safe load capacity of each.
(f) Building cross sections as required to show construction, arrangement, style, makeup, and span of runway track beam sections; and headroom and clearance distances as applicable, from the top of the bottom flange or the center of the runway track beams to columns, conduits, ducts, overhead building construction, light fixtures, runway conductors, and other obstructions.
(g) Plans and elevations delineating trolley and bridge travel range(s); locations of bridge, pendant, auxiliary, and

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fixed controls; the location, reach, and method of attachment of strain leads; the location of the crane in relation to other cranes; hook approach limits toward the sides and ends of the crane service area; and the hooks and operating ranges of other cranes as applicable.

(h) Location of transfer points and the location and reach of interlock control handles or operating pushbuttons.

(i) Indoor and outdoor service applications where weather protection is required, motor enclosure and any special protective design characteristics, and the location of wind lock seats.

(j) Any special limitation, such as design wheel loading, load concentration points, design wheel spacing, critical column loading, grade designation and characteristics of steel to be used, and runway girder reaction limitation.

(k) Bumper heights and sizes of bumper faces of existing cranes on the same runways when the crane is to extend or expand existing crane complexes.

(l) Kind of electric current, its voltage, phase, and frequency.

The building design considerations noted in CMAA No. 70 or 74 should be referred to in preparation of building plans and the schematic drawing.

* 6.8 Warranty. Procurement documents should contain the following:

"WARRANTY. The equipment including all components and parts, shall be warranted for a period of 1 year from the date of completion of the final acceptance tests, against defective materials, design, and workmanship. The warranty shall be extended by the crane manufacturer without recourse by the Government to suppliers of components or parts. Upon receipt of notice from the Government of failure of any part of the guaranteed equipment during the warranty period, the contractor shall furnish, at his own expense and reasonable time, all necessary replacement parts, delivered to the location of the equipment."

* 6.9 Changes in classification. Former style 1 and 2 is now identified as type I. Former style 3 and 4 is now identified as type II.

* 6.10 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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FORM A

Test of (Main) (Auxiliary) Hoist

No Load on Hook

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM B

Test of (Main) (Auxiliary) Hoist

25 Percent of Rated Capacity on Hook

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										
Alinement										
Lubrication										
Vibration										
Brake(s)										

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM C

Test of (Main) (Auxiliary) Hoist

50 Percent of Rated Capacity on Hook

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										
Alinement										
Lubrication										
Vibration										
Brake(s)										

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM D

Test of (Main) (Auxiliary) Hoist

75 Percent of Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM E

Test of (Main) (Auxiliary) Hoist

100 Percent of Rated Capacity on Hook

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM F

Trolley Travel Test

Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM G

Bridge Travel Test

Rated Capacity on Hook, Centered Between Runways

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM H

Test of (Main) (Auxiliary) Hoist

125 Percent of Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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3950-N142

DOCUMENT IDENTIFIER AND TITLE
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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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